YE ZHENG

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ABOUT ME

Over the past five years, I have primarily focused on the design and analysis of foundational algorithms for formal privacy and formal safety.

EXPERIENCE

Nanyang Technological University (Singapore) Jul 2026 – Present

- Research Fellow, working with Dr. Wei Dong
- Research Interests: Differential Privacy

EDUCATION

Rochester Institute of Technology (Rochester, USA) Sep 2023 – Oct 2026

- Ph.D. in Computing and Information Sciences, advised by Dr. Yidan Hu
- Research Interests: AI Privacy, Differential Privacy (Formal Privacy)

Shenzhen University (Shenzhen, China) Sep 2020 – Jun 2023

- M.S. in Software Engineering, advised by Dr. Jiaxiang Liu
- Research Interests: Neural Network Verification (Formal Verification)

Henan University (Kaifeng, China) Sep 2016 – Jun 2020

- B.S. in Mathematics, advised by Dr. Zhonghua Wang
- Major: Pure Mathematics

PUBLICATIONS (1st-author then co-author; full list at Google Scholar)

Conference Publications:

1. **[PETS'26]** AUDAGENT: Automated Auditing of Privacy Policy Compliance in AI Agents
   Ye Zheng, Yimin Chen, and Yidan Hu
2. **[PETS'26]** Quantifying Classifier Utility under Local Differential Privacy
   Ye Zheng and Yidan Hu
   ★ **Andreas Pfitzmann Best Student Paper Award**
3. **[PETS'26]** TraCS: Trajectory Collection in Continuous Space under Local Differential Privacy
   Ye Zheng and Yidan Hu
4. **[PETS'25]** Optimal Piecewise-based Mechanism for Collecting Bounded Numerical Data under Local Differential Privacy
   Ye Zheng, Sumita Mishra, and Yidan Hu
   ★ **Artifact Award Runner-up**
5. **[PETS'25]** Locally Differentially Private Frequency Estimation via Joint Randomized Response
   Ye Zheng, Shafizur Rahman Seeam, Yidan Hu, Rui Zhang, and Yanchao Zhang

6. **[FSE'22 Demonstrations]** MpBP: Verifying Robustness of Neural Networks with Multi-path Bound Propagation
   Ye Zheng, Jiaxiang Liu, and Xiaomu Shi
7. **[JOS'22]** (in Chinese) Multi-path Back-propagation Method for Neural Network Verification
   Ye Zheng, Xiaomu Shi, and Jiaxiang Liu
8. **[ICDCS'26]** PrivAR: Client-Side Privacy Framework for Real-Time Location-Based Augmented Reality
   Shafizur Rahman Seeam, Ye Zheng, Zhengxiong Li, and Yidan Hu
9. **[PETS'26]** Frequency Estimation of Correlated Multi-attribute Data under Local Differential Privacy
   Shafizur Rahman Seeam, Ye Zheng, and Yidan Hu
10. **[CNS'24]** Multi-sensor Data Privacy Protection with Adaptive Privacy Budget for IoT Systems
    Xinyi Liu, Ye Zheng, Zhengxiong Li, and Yidan Hu
11. **[SAS'23]** Boosting Multi-neuron Convex Relaxation for Neural Network Verification
    Xuezhou Tang, Ye Zheng, and Jiaxiang Liu

Selected Awards

Andreas Pfitzmann Best Student Paper Award, PETS Award Committee — Apr 2026

Artifact Award Runner-up, PETS Artifact Evaluation Committee — Aug 2025

Outstanding Graduate, Shenzhen University — Jun 2023

National Scholarship, Ministry of Education, China — Sep 2022

Software Projects

TypeNext — LLM-powered inline auto-completion in any APP, e.g. Chrome, Office, Typora, etc. Supports local models and remote APIs.

Academic Services

Reviewer for: TKDE (2027), PETS (2027), TOIT (2026), TKDD (2026, 2025), TASE (2024), SAS (2024)